

Finish the Hardware

44 steps, from soldered headers to a working device on your face

Good news: the headers change everything for the better. Every board connection is now **push-on**. Only ONE joint still needs the iron — the touch disc, step 19. Everything else plugs together and can be unplugged again when you're debugging.

One thing to know before you print pods. Headers plus a pushed-on jumper socket add about 10mm to the board's height. The pod's vertical size follows that — a bare board gives a **10mm** pod; with headers it's **19mm**. That's the bulk the two-pod split existed to avoid. You don't have to decide yet: step 39 lays out the choice, and both files are ready.

THE SIX PARTS

Steps	Part	Iron needed?
1-8	Check the headers you just soldered	Only if you find a bridge
9-17	Plug in the amp and speaker	No
18-21	Make the touch pad	Yes — one joint
22-28	First power-on and flash the firmware	No
29-37	Start the server, test the camera	No
38-44	Test audio, then into the pods	No

Same rules as before. Do one step. Check the green line actually happened. Only then move on. If the green line didn't happen, the red line tells you what to do.

PART 1 — Check what you just soldered

Steps 1–8. Before anything gets powered.

1 Make sure the USB cable is unplugged. ☐

You should see: nothing connected to the board.

2 Lower your eye to table level and look along the row of 7 pins. ☐

From the side, like looking along a fence — not down from above.

You should see: 7 separate solder cones with clear gaps between them.

If two are joined: wipe the iron tip clean, drag it firmly across the join and off the edge of the board. Look again.

3 Do the same on the other row of 7. ☐

You should see: 7 more separate cones.

4 Set the multimeter to beep/continuity and touch the probes together. ☐

The dial position marked ")))" or a diode triangle.

You should see: it beeps when they touch, silence when apart.

5 Touch one probe to the 3V3 pin, the other to the GND pin. ☐

Touch the metal pins themselves, not the plastic.

You should see: SILENCE.

If it beeps: STOP. Those two are joined by stray solder — a dead short that can kill the board the moment you plug in. Find it and drag it off. Do not continue until this is silent.

6 Test every neighbouring pair on both rows. ☐

5V – GND , GND – 3V3 , 3V3 – D10 , D10 – D9 , D9 – D8 , D8 – D7 , then the same down the D0 – D6 row.

You should see: silence every time. That's 12 tests.

Any beep = a bridge. Same fix. Worth the two minutes — this is the single most common way a soldered board gets destroyed.

7 Check the Sense expansion board is still firmly attached underneath. ☐

The smaller board with the camera socket on it. Press it gently — it should be flush with no gap at either end.

You should see: the two boards sitting tight together.

If it's loose: press it back on. Heat from soldering can loosen it.

8 Check the camera ribbon is still latched in its connector. ☐

The dark flip-latch should be pressed flat. Tug the ribbon very gently — it must not move.

You should see: a seated ribbon, latch closed.

If it moved: lift the latch, push the ribbon fully in with the contacts facing down, press the latch flat.

PART 2 — Plug in the amp and speaker

Steps 9–17. No soldering.

9 Find five jumper wires with a plastic SOCKET on both ends. ☐

"Female-to-female" — a small plastic block with a hole at each end. Your kit has 20. You want red, black, yellow, green, blue if possible.

You should see: five wires, socket at both ends.

Only male-to-male (bare pins)? Those won't work on header pins. Use female-to-female, or female-to-male with the socket end on the board.

10 Look at your amplifier board and find its pin labels. ☐

You want to be able to point at **VIN** (or **5V**), **GND**, **BCLK**, **LRC** and **DIN**.

You should see: five named pins you can point to.

Different names? **BCLK** = **SCK** = **CLK** · **LRC** = **WS** = **LRCLK** · **DIN** = **DATA**. Same things.

11 Push the BLACK wire onto the board's GND pin, other end onto the amp's GND. ☐

Push straight down until it's fully seated. It should need a small amount of force.

You should see: the socket fully covering the pin, no metal showing.

12 RED wire: board 3V3 → amp VIN. ☐

You should see: two wires connected.

Amp says 5V only? Use the board's **5V** pin instead — but note it only carries power while USB is plugged in.

13 YELLOW wire: board D8 → amp BCLK. ☐

You should see: three wires connected.

Note: **D8** is the pad label printed on the board. It is not GPIO 8. Go by the printed label.

14 GREEN wire: board D9 → amp LRC. ☐

You should see: four wires connected.

15 BLUE wire: board D10 → amp DIN. ☐

You should see: five wires connected. The whole audio link is done.

16 Connect the speaker's two wires to the amp's two output pins. ☐

Labelled **OUT+** / **OUT-** or just **+** / **-**. Screw terminals: strip 5mm, insert, tighten. Pins: push jumpers on. Bare pads: this is the one place you may need the iron.

You should see: the speaker attached to the amp.

Which wire to which? Doesn't matter. A single speaker has no correct way round.

17 Leave GAIN, SD and SHUTDOWN completely unconnected. ☐

If your amp has these extra pins, ignore them.

You should see: nothing plugged into them.

Why it matters: **SD** usually means *shutdown*. Connecting it to ground silences the amp completely, and you will spend an hour hunting a fault that isn't there.

PART 3 — The touch pad · PART 4 — Power and flash

Steps 18–28. One solder joint, then software.

- 18

Rub sandpaper over one face of a coin until it looks dull and scratched. ☐

Any coin, or a wide screw head. Scuff an area about the size of a pea.
You should see: a dull scratched patch, not shiny.
Why: solder will not stick to a shiny coin. This step is the whole reason the touch pad works.
- 19

Solder one end of a jumper wire to the scuffed patch. The only joint left. ☐

Use a female-to-male jumper and solder the **male (pin) end** to the coin, so the socket end is free to push onto the board. A coin is a big lump of metal — hold the iron on it 5–10 seconds, not 2.
You should see: the wire firmly attached to the coin.
If solder won't stick: not scuffed enough, or not hot enough. Scuff harder, hold longer.
- 20

Push the coin wire's free end onto the board pin labelled `D2`. ☐

Left-hand row, third pin down from the USB end.
You should see: the coin connected to `D2` and nothing else.
- 21

Re-test `3V3` to `GND` with the meter, one last time. ☐

Everything is now connected, so this catches anything the wiring introduced.
You should see: silence.
If it beeps now but didn't at step 5: the amp or a jumper is the cause. Unplug the amp and test again.
- 22

Plug in the USB cable. ☐

You should see: a small LED on the board light up.
Nothing? Many USB cables are charge-only with no data wires. Try a different cable before assuming the board is faulty.
- 23

Within 5 seconds, dab a fingertip on the board and on the amp. ☐

You should see: both cool or barely warm.
If anything is hot: UNPLUG NOW. You have a short. Go back to step 5.
- 24

Open the `firmware` folder in VS Code with the PlatformIO extension. ☐

You should see: PlatformIO recognises the project and shows a build toolbar at the bottom.
- 25

Copy `src/secrets.h.example` to a new file called `src/secrets.h`. ☐

You should see: both files in the `src` folder.
- 26

Put your WiFi name and password into `secrets.h`. Leave the IP for now. ☐

You should see: `WIFI_SSID` and `WIFI_PASSWORD` filled in.
Note: the ESP32 only works on **2.4GHz** WiFi. If your network is 5GHz-only it will never connect.
- 27

Press Build (the tick icon). ☐

You should see: `SUCCESS` in the terminal. First build downloads a lot and takes several minutes.
Errors? Send me the last 20 lines. Don't start changing code.
- 28

Press Upload (the arrow icon). ☐

You should see: a progress bar, then `SUCCESS`.

PART 5 — Server and camera

Steps 29–37. Camera only. No audio yet.

Camera first, audio second — never both at once. If both are on and it fails, you can't tell which half broke.

- 29 Open a terminal in the `server` folder and install what it needs. ☐

```
cd server
pip install -r requirements.txt
```

You should see: packages install, ending without an error.

pyzbar fails? Needs the system `zbar` library — Mac: `brew install zbar`.

- 30 Create a file called `.env` in the `server` folder with your API keys. ☐

You should see: a `.env` file. It's gitignored, so it won't be committed.

- 31 Start the server. ☐

```
uvicorn main:app --host 0.0.0.0 --port 8000
```

You should see: `Uvicorn running on http://0.0.0.0:8000`. Leave this terminal open.

- 32 In a browser, open `http://localhost:8000/health`. ☐

You should see: a small JSON response, not an error page.

- 33 Test the AI with no hardware at all — photograph a medicine box with your phone, copy it to your laptop, and run: ☐

```
curl -X POST http://localhost:8000/read_label \
-F "image=@box.jpg"
```

You should see: the server describe what's on the box.

The most valuable test in the project — it proves the AI half works with no wiring involved. Try ten real products.

PART 5 continued — point the board at your laptop

Steps 34–37.

- 34

Find your laptop's IP address on the local network.

Mac `ipconfig getifaddr en0` · Windows `ipconfig` (IPv4) · Linux `hostname -I`
You should see: something like `192.168.x.x` . **Write it down.**
- 35

Put that IP into `secrets.h` as `SERVER_HOST` , then Upload again.

You should see: another `SUCCESS` .
Laptop and board must be on the SAME WiFi network. Not guest network, not phone hotspot on one and home WiFi on the other.
- 36

Open the Serial Monitor (the plug icon) and press RESET on the board.

You should see: `Connecting to WiFi` then `WiFi connected` .
Says FAILED? Check the password, and check it's a 2.4GHz network.
- 37

Touch the coin. Watch BOTH the serial monitor and the server terminal.

You should see: the board report a capture, and the server log a request arriving.
Camera fails? PSRAM, not wiring — fixed in `platformio.ini` , so rebuild after pulling. **Nothing on touch?** Try holding the coin rather than tapping.

PART 6 — Audio, then onto your face

Steps 38–44.

38 Touch the coin again and listen to the speaker. ☐

You should see (hear): a voice reading out whatever the camera is pointed at. **That's the whole product working.**

Silence? Check **SD / SHUTDOWN** is unconnected (step 17). Then meter amp **VIN** → **GND**, expect ~3.3V.

Static or buzzing? Yellow and green are swapped. **D8** → BCLK, **D9** → LRC.

Very quiet? Try the board's **5V** pin instead of **3V3**, if your amp is rated for it.

39 Decide which pod plate to print. ☐

Keeping the headers → print **FINAL_petg_plate_HEADERS.stl**. Pods are 19mm tall. Stick out more, but nothing to undo.

Removing the headers → print **FINAL_petg_plate.stl**. Pods are 10mm tall and sit much closer to the head, but you'd desolder the headers and solder wires flat to the pads.

You should see: one file chosen.

My honest suggestion: print the HEADERS version now. A working chunky prototype beats an elegant one that isn't finished. Slimming it down is a week-4 job if you have time.

40 Print your chosen plate in PETG, and **FINAL_tpu_rings.stl** in TPU. ☐

Layer height 0.16mm, 4 perimeters. The rings are tiny — about 10 minutes each.

You should see: two pod bases, two lids, and four rings.

41 Slide two rings onto one temple arm of your glasses, from the ear end. ☐

They're deliberately undersized — they should need a firm stretch to go on.

You should see: rings that grip and don't slide when pushed.

Too loose? Reprint with a SMALLER opening — not larger. Reduce **RING_IN_OUT** by 0.5mm in the **.scad**.

Won't stretch on? Reduce **RING_WALL** to 1.0mm — a thinner wall stretches far more easily.

42 Put the board and camera into the front pod, amp and speaker into the rear pod. No glue yet. ☐

Camera lens lined up with the hole. Speaker under the grille holes.

You should see: both lids close flat.

A lid won't close? Something is taller than the CAD expects. Measure it and tell me the number rather than forcing the lid.

43 Slide each pod onto its rings, along the arm. ☐

The slot slides onto the rail from the end — it won't push on from above. That's the point.

You should see: pods that sit firmly and can't be pulled straight off.

44 Put the glasses on and wear them for ten minutes. ☐

You should see: nothing pressing into your head, arm not splayed, pods not sliding.

Something digs in? Tell me where. That's a CAD fix and it's a quick one.

That's the hardware done. Everything after this is reliability, the wake word, and making it look finished.